PAPER ID-310526

Subject Code: BCE501

Roll No: | | | | | | | | | | | | | |

BTECH

(SEM V) THEORY EXAMINATION 2025-26

GEOTECHNICAL ENGINEERING

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define thixotropy& sensitivity.	1	K2
b.	What are the different types of minerals are present in the soil also mention it's functions.	1	K1
c.	State Darcy's law.	2	K1
d.	Define quick sand condition.	2	K2
e.	What are the governing factors of compaction of soil?	3	K1
f.	Define shear strength of the soil.	4	
g.	Define stability number.	5	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the various process of soil formation.	1	K2
b.	Explain the various factors affecting the permeability of a soil.	2	K1
c.	Explain the differences between Compaction & Consolidation.	3	K2
d.	A concentrated load of 22.5KN acts on the surface of a homogeneous soil mass of large extent. Find the stress intensity at a depth of 15m and directly under the load and also at a horizontal distance of 7.5m. Use Boussinesq's equation.		
e.	State the assumptions of the Rankine's earth pressure theory.	5	K1

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	The dry density of sand with porosity of 0.387 is 1600 kg/m^3. Find the void ratio of the soil and the specific gravity of soil solids. (Take unit weight of water is 1000kg/m^3)	1	K2
b.	Explain the index properties of the soil.	1	K1

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Derive a relation for the permeability of stratified soil deposits when flow is perpendicular to the bedding plane and when the flow is parallel to the bedding plane.	2	K2
b.	A sand sample of 35 cm^2 cross sectional area and 20 cm long was tested in a constant head permeameter. Under a head of 60 cm, the discharge was 120 ml in 6 min. the dry weight of sand used for the test was 1120 gm and specific gravity of soil is 2.68. Determine: 1. It's the hydraulic conductivity in cm/ sec. 2. Discharge velocity. 3. Seepage velocity.	2	K2

QP26DP1_311 | 31-Dec-2025 9:01:37 AM | 49.42.35.22

QP26DP1_311 | 31-Dec-2025 9:01:37 AM | 49.42.35.22

PAPER ID-310526

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM V) THEORY EXAMINATION 2025-26
GEOTECHNICAL ENGINEERING

TIME: 3 HRS **M.MARKS: 70**

5. Attempt any *one* part of the following: 07 x 1 = 07

a.	State the assumptions of the Terzaghi's theory of one-dimensional consolidation.	3	K2
b.	In a consolidation test the following results have been obtained. When the load was changed from 50 KN/m^2 to 100 KN/m^2, the void ratio changed from 0.70 to 0.66. Determine the coefficient of compressibility(m_v) and the compression index (C_c).	3	K2

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Mention the different types of test which are available for the shear strength test. Explain triaxial shear test.	4	K2
b.	Calculate the potential shear strength on a horizontal plane at a depth of 3m below the surface in a formation of cohesionless soils when the water table is at a depth of 3.5 m. The degree of saturation may be taken as 0.5. Void ratio is 0.60, grain specific gravity is 2.70 and angle of internal friction is 30°. What will be the modified shear strength if the water table reachesthe ground surface?	4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	An excavation is to be made in a soil deposit with a slope of 25° to the horizontal and to a depth of 25m. the soil has the following properties: 1. Effective cohesion is 35 KN/m^2, 2. Effective angle of internal friction is 15°, 3. Bulk unit weight is 20 KN/m^3. Determine the factor of safety of the slope assuming full friction is mobilized and if the factor of safety with respect to cohesion is 1.5, what would be the factor of safety with the respect to friction?	5	K2
b.	Mention the difference between finite slopes and infinite slopes. Explain Bishop Method for slope analysis.	5	K1

QP26DP1_311 | 31-Dec-2025 9:01:37 AM | 49.42.35.22

QP26DP1_311 | 31-Dec-2025 9:01:37 AM | 49.42.35.22